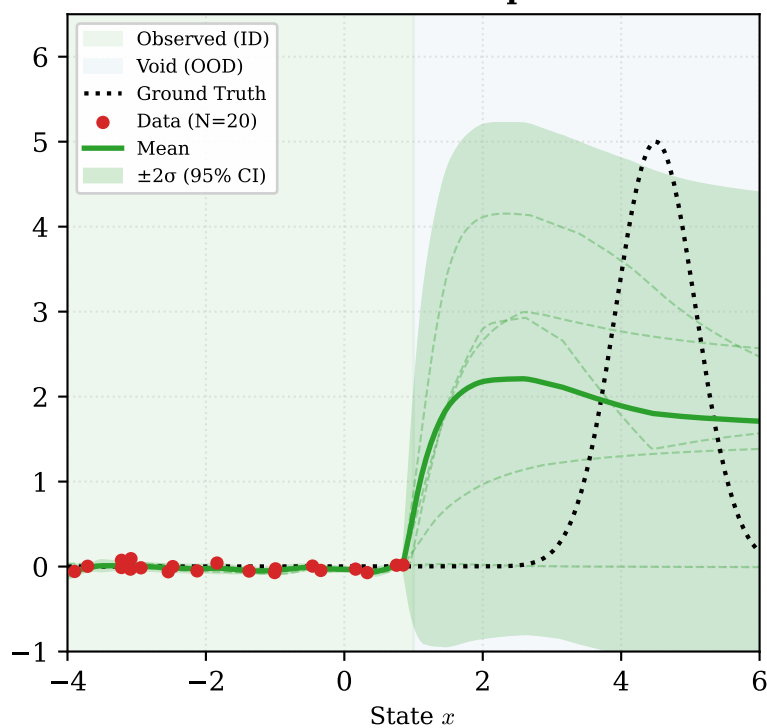
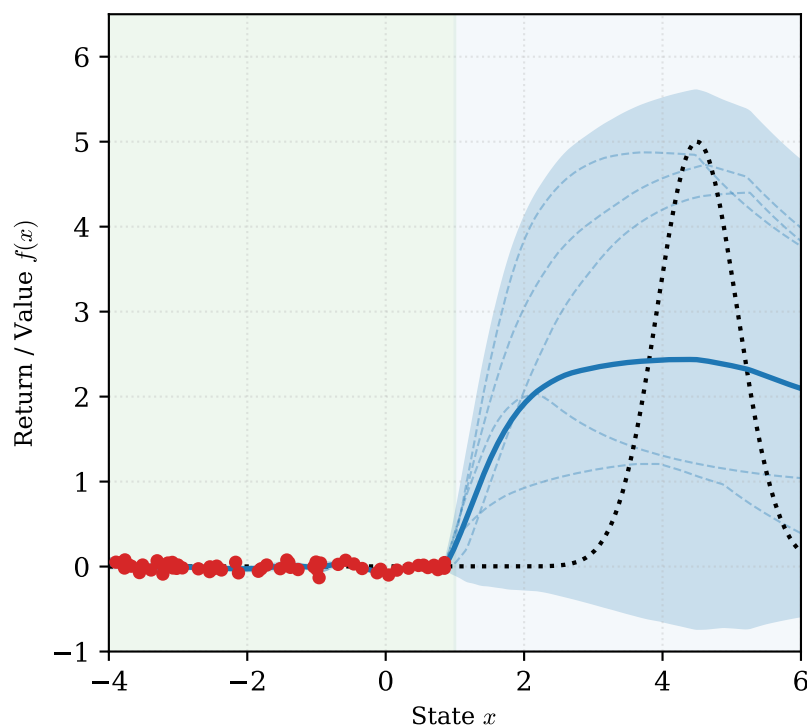


# Posterior Contraction vs. Epistemic Exploration — DP-BNN Resolves the Fundamental Bayesian Trade-Off

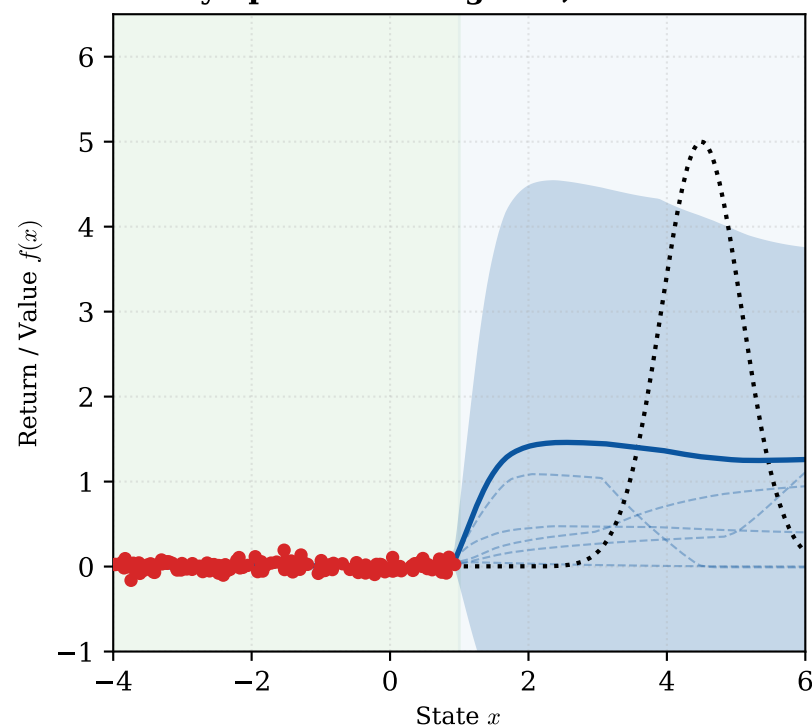
**(a) DP-BNN: Small Data ( $N=20$ )**  
Broad in-between exploration



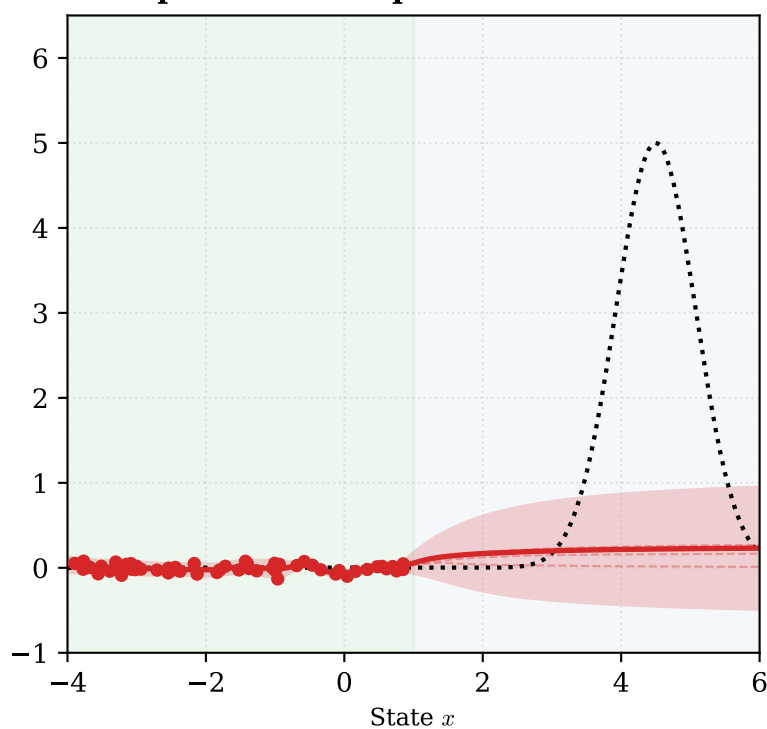
**(b) DP-BNN: Moderate Data ( $N=60$ )**  
In-distribution variance contracts to 0



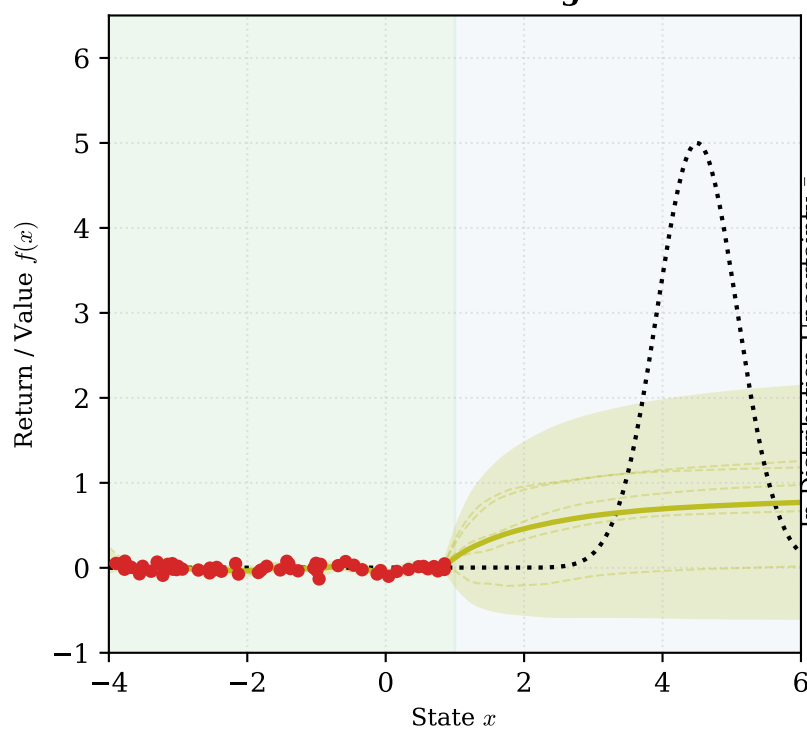
**(c) DP-BNN: Dense Data ( $N=150$ )**  
Asymptotic convergence; void intact



**(d) Deep Ensembles ( $N=60$ )**  
Epistemic Collapse:  $\sigma \approx 0.05$  in void



**(e) BootDQN + Priors ( $N=60$ )**  
Unbounded: shoots into negative returns



**(f) Posterior Contraction Rate  $\mathcal{O}(1/\sqrt{N})$**

